What Is a Computer Program

A set of instructions that tells a computer what to do.

1 Definition

A **computer program** is a collection of ordered instructions written in a programming language.

These instructions tell the computer how to perform a task, solve a problem, or automate a process.



2 Everyday Examples



Calculator app
➡ performs calculations



Web browser
➡ opens and displays websites

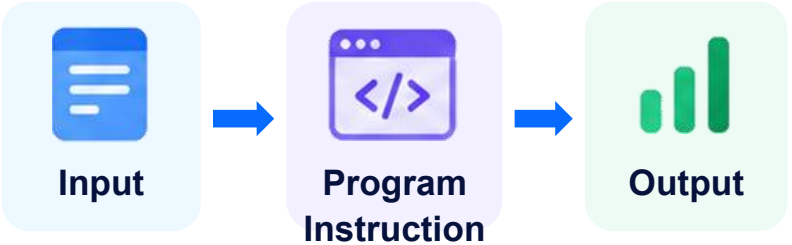


Game
➡ responds to player actions



ATM software
➡ processes banking operations

3 How It Works



1. The user or system provides input.
2. The program processes the input using rules and logic.
3. The computer produces output or a result.



Key idea: Hardware is the physical machine, while software (including programs) gives the machine instructions.

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What Is Programming

The process of designing instructions for a computer.

1 Definition

Programming is the process of writing, organizing, testing, and maintaining instructions that a computer can follow.

It turns ideas and problem-solving steps into working software.



2 What programmers do

- **Analyze a problem**
- **Design a solution**
- **Write source code**
- **Test and improve the program**

3 Simple workflow

Problem → Algorithm / Plan → Code → Test → Result

1. Programming is not only typing code.
2. It also includes logic, debugging, and continuous improvement.

Key idea: A programmer uses logic and creativity to build software that solves real problems.

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What Is a Programming Language

A formal way to communicate instructions to a computer.

1 Definition

A **programming language** is a set of words, symbols, and rules used to write computer programs.

It allows humans to express logic in a form that computers can process.



2 Why We Need Programming Languages

Readable for humans

Use familiar words and clear structure to express ideas.

Structured and precise

Follow defined rules to avoid ambiguity.

Helps create software efficiently

Provides tools and features to build programs faster.

Can be translated into machine-executable form

Converted into instructions that computers can run.

3 Examples

- **JavaScript** — web interactivity
- **Python** — automation and data
- **C#** — desktop, web, and enterprise apps
- **Java** — cross-platform applications
- **SQL** — working with databases

Human logic → **Programming language** → **Computer execution**

Key idea: Different languages have different strengths, but all are used to describe instructions clearly.

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Classification of Programming Languages

Languages can be grouped in different useful ways.

1 By Level

Low-level
close to hardware
(Machine language, Assembly)

High-level
easier for humans to read and write
(JavaScript, Python, C#)



2 By Execution Method

Compiled
translated
before running

Interpreted
executed step by step
or by an engine

Hybrid / VM-based
uses bytecode or runtime environments

3 By Purpose

General-purpose
many kinds
of software

Domain-specific
focused on a particular area, such as SQL for databases

4 By Programming Style

Procedural

Object-oriented

Functional

Event-driven

Many languages support more than one style.

Key idea: A language may belong to several classifications at the same time.

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What Is a Programming Paradigm

A general style or way of thinking about programming.

1 Definition

A **programming paradigm** is a model or approach for structuring programs and solving problems.

It influences how code is organized and how developers think about logic.



2 Common Paradigms

Procedural
organizes code as steps and procedures

Object-Oriented
organizes code around objects and classes

Functional
focuses on functions and data transformations

Event-Driven
reacts to user actions or events

3 Why Paradigms Matter

1. They shape program design.
2. They affect readability and maintenance.
3. A single language may support multiple paradigms.

Example:
JavaScript supports procedural, object-oriented, and event-driven programming.

Key idea: A paradigm is not a language; it is a way of writing and organizing code.

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How Does Source Code Run

From human-written code to actual execution.

1 Definition

Source code is the human-readable code written by a programmer.

To run, it must be processed by a compiler, interpreter, or runtime engine.



2 From Code to Execution

Source Code → Translator / Engine → Machine Instructions → Execution → Output

Examples: compiler, interpreter, JavaScript engine

3 JavaScript Example

JavaScript file → Browser or Node.js → JS Engine parses and executes → Console output / page behavior

1. The engine reads the code
2. It converts or optimizes it internally
3. The computer executes the instructions

Key idea: Computers do not directly understand high-level source code; a runtime system bridges the gap.

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What Is an IDE

An integrated environment for writing and managing code.

1 Definition

IDE stands for **Integrated Development Environment**.

It is software that helps developers write, organize, run, and debug code in one place.



2 Common IDE features

- **Code editor**
- **Syntax highlighting and auto-completion**
- **Run and debug tools**
- **Project/file management**
- **Extensions or built-in tools**

3 Why it is useful

- **Improves productivity**
- **Reduces errors**
- **Keeps tools in one workspace**
- **Makes development easier for beginners**

The IDE workflow

Edit → **Run** → **Debug** → **Manage files** = **IDE workflow**

Example: Visual Studio is a full IDE, while Visual Studio Code is a lightweight editor that can become IDE-like with extensions.

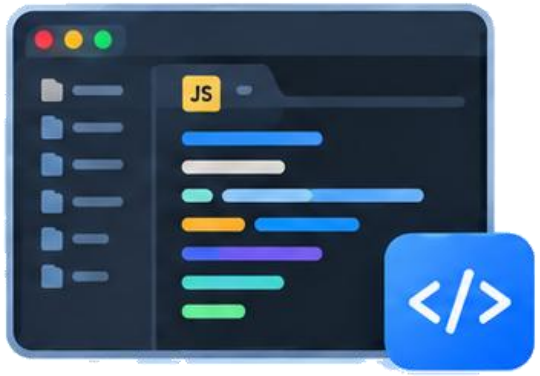
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Our Development Tools for JavaScript Development

The main tools we will use in this course.

1 Visual Studio Code

- Our main code editor
- Used to write and organize JavaScript files



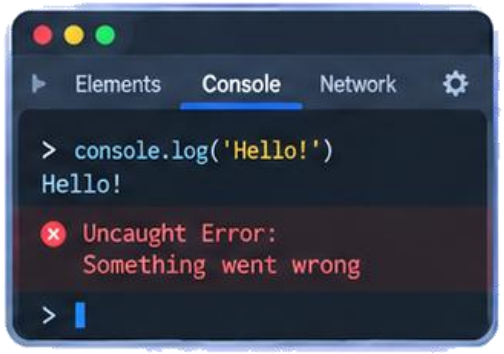
2 Web Browser

- Runs JavaScript in web pages
- Chrome or Edge can be used for testing



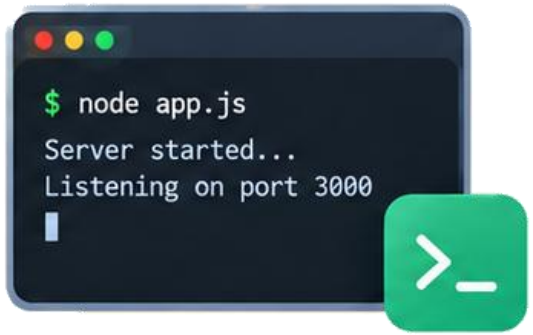
3 Browser Developer Tools

- Console for output and errors
- Inspect elements, network activity, and debugging



4 Node.js and Terminal

- Runs JavaScript outside the browser
- Useful for command-line execution and tools



Write code in VS Code → **Run in Browser or Node.js** → **Inspect with DevTools / Console**

Key idea: Good tools make learning faster, testing easier, and debugging clearer.

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Your First JavaScript Program

A simple start with JavaScript output.

1 Goal

We will write a simple program that prints a message.

This is often the first step when learning a new language.



2 Code example

```
console.log("Hello, World!");
```

JavaScript syntax

3 Result and explanation

```
Hello, World!
```

1. console.log() sends text to the console
2. Text inside quotation marks is a string
3. The semicolon marks the end of the statement

How to run it

Open the browser console or run the file with Node.js.

Key idea: Even a tiny program helps you understand syntax, output, and how code is executed.

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